

Chapter 1E: Variant, Microvascular, and Nonobstructive Angina

Medvale Internal Medicine - Cardiovascular Medicine

Source-use note. This is original educational writing. The uploaded cardiology material is used as a coverage checklist, especially for variant angina, which emphasizes transient coronary vasospasm, rest/night pain, transient ST elevation, angiographic provocation concepts, and treatment with calcium channel blockers and nitrates. Current guideline sources are used only for broad clinical framing of chronic coronary disease, INOCA, and ACS/MINOCA concepts.

Why this matters clinically

Not every patient with ischemic chest pain has a fixed epicardial coronary stenosis, and not every patient with a positive troponin has the classic plaque-rupture myocardial infarction pattern. This chapter covers the gray zone after stable angina, acute coronary syndrome, and myocardial infarction: patients with angina from transient vasospasm, ischemia from small-vessel dysfunction, and infarct-like presentations without obstructive coronary disease.

These entities are important because they can look like familiar coronary syndromes while requiring different reasoning. A patient with variant angina can have severe rest chest pain and transient ST elevation because an epicardial artery spasms. A patient with microvascular angina can have exertional ischemic symptoms and abnormal stress imaging even when angiography shows no obstructive epicardial coronary artery disease. A patient with MINOCA can meet criteria for myocardial infarction while angiography shows no stenosis of 50% or greater, forcing the clinician to search for vasospasm, embolism, spontaneous coronary artery dissection, myocarditis mimics, Takotsubo syndrome, or supply-demand injury.

For graph-RAG purposes, the key association is that ischemic symptoms should be connected to mechanism. Chest pain can arise from fixed plaque, acute thrombus, transient spasm, endothelial dysfunction, microvascular dysfunction, myocardial inflammation, stress cardiomyopathy, coronary embolism, or dissection. The management changes when the mechanism changes.

Numbers, cutoffs, and timelines

Variant angina often occurs at rest and classically appears at night or in the early morning. During symptoms, the ECG may show transient ST elevation, which can normalize after the spasm resolves. MINOCA frameworks commonly use the absence of obstructive coronary disease, often defined as no epicardial coronary stenosis of 50% or greater, in a patient who otherwise meets criteria for myocardial infarction. Troponin must show an acute rise and fall above the 99th percentile to support myocardial infarction rather than chronic injury.

Microvascular angina is supported by evidence of ischemia with nonobstructive epicardial coronaries and may be further characterized by reduced coronary flow reserve, elevated microvascular resistance, or microvascular spasm during specialized testing. Takotsubo syndrome often shows transient left ventricular systolic dysfunction extending beyond one coronary territory. SCAD is classically associated with younger or middle-aged women, the postpartum period, fibromuscular dysplasia, connective tissue disease, and emotional or physical stress.

Topic	Number or pattern	Clinical use
MINOCA anatomy	No stenosis $\geq 50\%$	MI syndrome without obstructive culprit lesion.
Variant angina ECG	Transient ST elevation	Spasm can briefly cause transmural ischemia.
Troponin in MINOCA	Rise/fall above 99th percentile	Confirms acute myocardial injury pattern.

Topic	Number or pattern	Clinical use
Dressler timing	Weeks to months	Distinguishes immune pericarditis from recurrent ACS.

Core illness scripts

Syndrome	Mechanism	Clinical pattern
Variant angina	Epicardial coronary vasospasm	Rest/nocturnal pain; transient ST elevation.
Microvascular angina	Small-vessel dysfunction	Angina/ischemia with nonobstructive coronaries.
MINOCA	MI with no obstructive culprit	Troponin rise/fall plus ischemia; mechanism search required.
SCAD	Intramural hematoma or dissection	ACS in younger/postpartum woman or low-risk patient.
Takotsubo	Stress cardiomyopathy	Troponin/ECG mimic; wall motion beyond one artery.

Variant angina: the vasospasm model

Variant angina, also called Prinzmetal angina, is ischemic chest pain caused by transient coronary artery spasm. The spasm is usually in an epicardial coronary artery and can occur in a segment that already has atherosclerosis or in a vessel that appears relatively normal between attacks. The defining idea is that the artery narrows dynamically rather than being fixed at a constant stenotic diameter. This is why the patient may have dramatic symptoms during an episode and a normal or near-normal evaluation afterward.

Unlike classic stable angina, variant angina often occurs at rest rather than predictably with exertion. It classically appears at night or in the early morning. During the painful episode, the ECG can show transient ST-segment elevation because the spasm can produce brief transmural ischemia. When the spasm relaxes, the pain and ST elevation may resolve. This transient pattern can mimic STEMI, but the episode may be recurrent, brief, and angiographically related to spasm rather than persistent thrombotic occlusion.

Common triggers and associations include cigarette smoking, cocaine or amphetamine use, emotional stress, cold exposure, hyperventilation, and vasoconstrictive medications. Some patients have other vasospastic conditions such as Raynaud phenomenon or migraine. The mechanism involves vascular smooth muscle hyperreactivity, endothelial dysfunction, autonomic imbalance, oxidative stress, and sometimes local inflammation around plaque. A patient does not need a severe fixed stenosis to have severe ischemia during spasm.

Clinical anchor: Rest chest pain + transient ST elevation + improvement with nitrates or calcium channel blockers = think coronary vasospasm. The uploaded cardiology source specifically emphasizes transient coronary vasospasm, episodes at rest/night, transient ST elevation during chest pain, angiographic spasm concepts, and vasodilator therapy with CCBs and nitrates.

Diagnosing variant angina

The most useful diagnostic clue is capturing the ECG during symptoms. A patient whose ECG is normal between episodes may show ST elevation, ST depression, or peaked T-wave changes only during pain. Because the abnormality is transient, a single normal ECG in clinic does not exclude the diagnosis. Ambulatory ECG monitoring can help when episodes are frequent, especially if nocturnal pain correlates with transient ischemic ST changes.

Coronary angiography may show obstructive CAD, nonobstructive CAD, or normal-appearing vessels. If the diagnosis remains uncertain, specialized provocative testing with acetylcholine or ergonovine can induce spasm under controlled conditions. This is not a casual screening test; it is performed in selected patients by experienced teams because provoked spasm can cause ischemia or arrhythmia.

Treating variant angina

Treatment is built around preventing spasm and removing triggers. Smoking cessation is crucial because smoking is strongly associated with coronary vasospasm. Cocaine, amphetamines, triptans, and other vasoconstrictive exposures should be stopped when relevant. The main anti-spasm medications are calcium channel blockers and nitrates. Dihydropyridine calcium channel blockers such as amlodipine reduce vascular smooth muscle contraction, while non-dihydropyridine agents such as diltiazem or verapamil can also reduce heart rate and demand. Long-acting nitrates can be added for persistent episodes, while sublingual nitroglycerin can relieve acute attacks.

Beta-blockers are not first-line for pure vasospastic angina and may worsen spasm in some settings, especially if nonselective beta-blockade leaves alpha-mediated vasoconstriction unopposed. This distinction matters because beta-blockers are often helpful for classic exertional angina, but vasospasm is not primarily a demand problem. The medication map should therefore connect stable angina to demand reduction and variant angina to vasodilation.

Microvascular angina: ischemia below the angiogram

Microvascular angina describes angina caused by dysfunction of the small coronary vessels rather than a large epicardial artery stenosis. Standard coronary angiography is excellent at showing the lumen of major coronary arteries, but it does not directly visualize the tiny resistance vessels that control myocardial perfusion at the tissue level. A patient can therefore have exertional ischemic symptoms, abnormal stress imaging, or impaired coronary flow reserve even when angiography does not show a flow-limiting epicardial lesion.

The pathophysiology is heterogeneous. Some patients have impaired endothelium-dependent vasodilation, meaning the microvasculature does not dilate appropriately when demand rises. Others have increased microvascular resistance, microvascular spasm, structural remodeling from hypertension or diabetes, inflammation, autonomic dysfunction, or heightened cardiac pain sensitivity. Because mechanisms differ, one medication does not work for every patient. This is why INOCA - ischemia with nonobstructive coronary arteries - is divided into endotypes such as vasospastic angina and microvascular angina.

Microvascular angina is particularly important in patients who continue to have angina despite "normal coronaries." Historically, these patients were sometimes reassured that nothing cardiac was wrong. That is unsafe reasoning. Nonobstructive coronary arteries do not necessarily mean noncardiac symptoms. The correct inference is narrower: there is no major obstructive epicardial stenosis on that angiogram.

Diagnostic clues for microvascular angina

Microvascular angina is suspected when a patient has angina-like symptoms, objective evidence of ischemia, and no obstructive epicardial coronary artery disease. Stress imaging may show inducible ischemia. Invasive coronary function testing may demonstrate abnormal coronary flow reserve, abnormal index of microcirculatory resistance, or microvascular spasm after acetylcholine. Noninvasive modalities such as PET or cardiac MRI can estimate myocardial blood flow or perfusion reserve in specialized settings.

A graph-RAG connection should link microvascular angina to diabetes, hypertension, LV hypertrophy, endothelial dysfunction, women with angina and nonobstructive coronaries, abnormal stress imaging, reduced coronary flow reserve, and persistent symptoms despite a clean angiogram. It should also link the phrase "normal coronary angiogram" to a warning: normal epicardial anatomy does not exclude microvascular ischemia.

Treatment logic for microvascular angina

Management starts with risk-factor control because microvascular dysfunction is often connected to hypertension, diabetes, dyslipidemia, obesity, smoking, inflammation, and sedentary lifestyle. Statins and ACE inhibitors or ARBs may improve endothelial function and reduce vascular risk in appropriate patients. Antianginal therapy is individualized. Beta-blockers can help when exertional symptoms are driven by heart

rate and demand. Calcium channel blockers may help when vasomotor dysfunction or microvascular spasm is suspected. Nitrates help some patients, but they are less predictably effective in microvascular angina than in classic epicardial spasm. Ranolazine may be considered for persistent symptoms, especially when blood pressure or heart rate limits other medications.

The treatment goal is not only to suppress chest pain but also to validate and phenotype the syndrome. Anxiety can coexist with ischemic syndromes, but it should not be used as a diagnostic shortcut when objective ischemia exists.

MINOCA: myocardial infarction with nonobstructive coronary arteries

MINOCA is not a final diagnosis. It is a working syndrome: the patient meets criteria for myocardial infarction, but coronary angiography does not reveal an obstructive coronary stenosis that explains the event. Many definitions use the absence of obstructive CAD at a threshold of 50% or greater diameter stenosis. The key point is that the patient has evidence of acute myocardial injury with ischemic features, yet the usual obstructive culprit lesion is not seen.

This creates a diagnostic trap. If the clinician stops after saying "MINOCA," the actual cause may be missed. MINOCA can result from plaque disruption that is not angiographically severe, coronary vasospasm, spontaneous coronary artery dissection, coronary embolism, microvascular dysfunction, supply-demand mismatch, or other coronary causes. It can also be mimicked by nonischemic myocardial injury such as myocarditis or Takotsubo syndrome. Therefore, MINOCA should trigger a structured search for mechanism.

Cardiac MRI is often valuable in MINOCA-style presentations because it can distinguish ischemic infarction patterns from myocarditis, Takotsubo syndrome, and other cardiomyopathies. Intravascular imaging such as OCT or IVUS can identify plaque rupture, plaque erosion, thrombus, or dissection that a standard angiogram may miss.

Mechanism	Clue	Management implication
Vasospasm	Rest pain, transient ST elevation	CCBs, nitrates, trigger avoidance.
Plaque disruption	ACS story, intracoronary imaging findings	ACS-style secondary prevention.
SCAD	Postpartum/young woman, long narrowing	Often conservative if stable.
Embolism	AFib, valve disease, endocarditis	Find source; anticoagulate when indicated.
Takotsubo	Stress trigger, diffuse wall motion pattern	Supportive cardiomyopathy care.
Myocarditis	Viral prodrome, CMR inflammation	Treat myocarditis complications.

Takotsubo syndrome: the stress cardiomyopathy mimic

Takotsubo syndrome, often called stress cardiomyopathy, can mimic MI with chest pain, ECG changes, and troponin elevation. It is often triggered by emotional stress, physical illness, neurologic injury, catecholamine excess, or severe systemic stress. The classic echocardiographic pattern is apical ballooning with transient left ventricular systolic dysfunction, but variants exist. The wall motion abnormality often extends beyond a single coronary artery territory, which helps separate Takotsubo from a typical focal coronary occlusion.

The mechanism is not fully explained by one pathway, but catecholamine-mediated myocardial stunning, microvascular dysfunction, and coronary vasomotor abnormalities are implicated. Troponin may be elevated, but the degree of LV dysfunction can seem disproportionate to the biomarker rise. Coronary angiography may show no obstructive culprit lesion. Management is supportive and tailored to complications such as heart failure, LV outflow tract obstruction, arrhythmia, or thrombus risk. Many patients recover systolic function over days to weeks, but acute complications can be serious.

Spontaneous coronary artery dissection

Spontaneous coronary artery dissection, or SCAD, is an important cause of ACS in patients who do not fit the usual atherosclerotic profile. It occurs when a tear or intramural hematoma forms within the coronary artery wall, compressing the true lumen and reducing blood flow. SCAD is classically associated with younger or middle-aged women, pregnancy or the postpartum period, fibromuscular dysplasia, connective tissue disorders, extreme emotional or physical stress, and sometimes no clear trigger.

SCAD can cause unstable angina, NSTEMI, STEMI, ventricular arrhythmia, or sudden death. It may appear angiographically as a dissection flap, contrast staining, long smooth narrowing, or abrupt vessel tapering. Management differs from routine plaque rupture because PCI can be technically challenging and may extend the dissection. Stable patients are often managed conservatively because many dissections heal spontaneously. Ongoing ischemia, left main involvement, hemodynamic instability, or high-risk anatomy may require revascularization.

Cocaine and stimulant-associated chest pain

Cocaine and amphetamines can cause chest pain through several overlapping mechanisms. They increase sympathetic tone, which raises heart rate, blood pressure, and myocardial oxygen demand. They also promote coronary vasoconstriction, platelet activation, thrombosis, endothelial dysfunction, and arrhythmia risk. A stimulant-exposed patient can therefore have simple tachycardia-related demand ischemia, variant-like vasospasm, plaque rupture ACS, myocarditis, aortic dissection, or malignant arrhythmias.

Treatment begins with the standard chest pain safety framework: ECG, troponins, vital signs, and evaluation for ACS or dissection when suggested. Benzodiazepines reduce sympathetic surge and are commonly used for acute cocaine toxicity. Nitrates and calcium channel blockers can relieve coronary vasospasm. Pure beta-blocker monotherapy during acute cocaine toxicity has traditionally been avoided because of concern for unopposed alpha-adrenergic vasoconstriction, though real-world practice is nuanced. For exam purposes, the high-yield association remains: cocaine chest pain connects to vasospasm, ACS, hypertension, tachycardia, aortic dissection risk, benzodiazepines, nitrates, and calcium channel blockers.

Medication distinctions across syndromes

Syndrome	Medication emphasis	Key caution
Stable exertional angina	Beta-blocker, nitrates, CCB, ranolazine	Match to HR, BP, EF.
Variant angina	CCBs and nitrates	Avoid beta-blocker monotherapy in pure spasm.
Microvascular angina	Risk-factor therapy plus tailored antianginals	Do not dismiss because angiogram is clean.
MINOCA	Treat identified mechanism	MINOCA is not a final diagnosis.
Stimulant chest pain	Benzodiazepines, nitrates, CCBs when indicated	Evaluate for ACS/dissection.

Diagnostic map: angina with nonobstructive coronaries

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Chest pain or ischemic syndrome
|
|-- Stable symptoms + objective ischemia + no obstructive CAD
|   -> INOCA: consider microvascular angina or vasospasm
|
|-- Troponin rise/fall + ischemic features + no stenosis >=50%
|   -> MINOCA working diagnosis
|   -> search for spasm, plaque disruption, SCAD, embolus,
|       myocarditis mimic, Takotsubo, or type 2 MI
|
|-- Rest/nocturnal pain + transient ST elevation
|   -> variant angina until proven otherwise
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High-yield exam associations

Stem clue	Best association
Rest chest pain at night + transient ST elevation	Variant / Prinzmetal angina
Angina + abnormal stress test + normal epicardial coronaries	Microvascular angina / INOCA
MI criteria + no stenosis >=50%	MINOCA working diagnosis

Stem clue	Best association
Postpartum NSTEMI with long smooth coronary narrowing	SCAD
Stress trigger + apical ballooning	Takotsubo syndrome
Cocaine chest pain + hypertension + tachycardia	Stimulant-associated vasospasm/ACS risk
Viral prodrome + troponin elevation	Myocarditis mimic

Common pitfalls

Pitfall 1: Equating normal coronary angiography with noncardiac pain. A normal or nonobstructive angiogram rules out major epicardial obstruction, not microvascular dysfunction, vasospasm, myocarditis, Takotsubo syndrome, or all cardiac causes of pain.

Pitfall 2: Treating vasospasm like ordinary demand angina. Beta-blockers reduce demand, but vasospasm is mainly a dynamic vasoconstriction problem. Calcium channel blockers and nitrates are central.

Pitfall 3: Stopping at the word MINOCA. MINOCA is a starting label. The clinician must search for the mechanism because treatment differs dramatically among plaque disruption, spasm, SCAD, embolism, myocarditis, and Takotsubo.

Pitfall 4: Dismissing symptoms as anxiety in patients with nonobstructive coronaries. Anxiety can amplify symptoms, but it should not erase objective evidence of ischemia or myocardial injury.

Practice questions

Question 1

A 46-year-old smoker has recurrent chest pain at 3 AM. ECG during pain shows transient ST elevation that resolves after nitroglycerin. Coronary angiography does not show severe fixed stenosis.

Answer: Variant / Prinzmetal angina. The rest/nocturnal timing, transient ST elevation, and nitrate response point to coronary vasospasm.

Question 2

A 58-year-old woman has exertional chest pressure and an abnormal stress perfusion study. Coronary angiography shows no obstructive epicardial CAD. Her symptoms persist despite reassurance.

Answer: Microvascular angina / INOCA. Nonobstructive epicardial coronaries do not exclude ischemia from coronary microvascular dysfunction.

Question 3

A 38-year-old postpartum woman presents with NSTEMI and angiography shows long smooth coronary narrowing suspicious for intramural hematoma.

Answer: Spontaneous coronary artery dissection. SCAD is an important ACS mechanism in younger or postpartum women.

Question 4

A patient has chest pain, troponin rise/fall, and ischemic ECG changes. Angiography shows no stenosis $\geq 50\%$.

Answer: MINOCA is a working diagnosis. The next step is to identify the mechanism.

Question 5

A patient develops chest pain after cocaine use with severe hypertension and tachycardia. ECG does not show STEMI.

Answer: Benzodiazepines reduce sympathetic surge; nitrates or calcium channel blockers can help vasospasm when indicated. Continue ACS evaluation.

Chapter summary

Variant angina, microvascular angina, and MINOCA separate ischemia from the oversimplified idea that all coronary symptoms require a fixed obstructive epicardial stenosis. Variant angina is caused by transient epicardial coronary spasm, often producing rest or nocturnal chest pain with transient ST elevation; treatment centers on calcium channel blockers, nitrates, and trigger avoidance. Microvascular angina is ischemia from coronary small-vessel dysfunction and can occur despite nonobstructive epicardial arteries; diagnosis depends on symptoms, objective ischemia, and tests of coronary flow or microvascular function when available. MINOCA is myocardial infarction with nonobstructive coronary arteries, often using a <50% stenosis framework, but it is not a final diagnosis. It requires a mechanism search for spasm, plaque disruption, SCAD, embolism, microvascular dysfunction, myocarditis mimic, Takotsubo syndrome, or supply-demand mismatch. The central lesson is mechanism-based treatment.

Selected sources used for coverage and clinical framing

1. Uploaded cardiology excerpt, especially cardiology2.pdf page 5, used as a coverage signal for variant angina, including transient coronary vasospasm, rest/night pain, transient ST elevation, angiography/provocation concepts, calcium channel blockers, nitrates, smoking cessation, and lipid risk-factor modification.
2. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA Guideline for Chronic Coronary Disease, used for modern chronic coronary disease, INOCA, antianginal, and mechanism-based management framing.
3. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for Acute Coronary Syndromes, used for contemporary ACS/MINOCA context and distinction between ACS syndromes and nonobstructive infarct presentations.